

Cascade PID Versus Output-Feedback LQR for the Fixed-Wing Flight Mode of a Hybrid VTOL UAV: Design, Robustness, and a Deterministic Hardware-in-the-Loop Comparison

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Source code and simulation data are available at <https://github.com/Kauemartinsxd/controle-pid-drone-hibrido>.

ABSTRACT This article compares two flight control laws — a classical cascade PID and an output-feedback linear-quadratic regulator (LQRy) — for the fixed-wing flight mode of a small hybrid vertical take-off and landing (VTOL) UAV, following the complete path from design to embedded validation. Both controllers are designed on a model linearized about the cruise trim of a nonlinear model identified from flight-test data, whose open-loop dynamics include an unstable spiral mode, and are evaluated against a common set of literature-grounded closed-loop requirements. In software-in-the-loop (SIL) simulation on the nonlinear model, the two laws are exercised under identical references on a 300-second combined mission; the requirements-retuned PID meets every requirement and is the tighter law on every guidance channel, while the deliberately soft fixed-gain LQRy is slower, concedes up to a factor of five in RMS tracking error, and violates the airspeed-overshoot requirement. A robustness study then inverts that verdict: under a severe lateral-directional model mismatch — an independent identification of the same airframe with sign-inverted lateral derivatives — the PID loses the mission outright, while the fixed-gain LQRy completes it with near-nominal error — a tolerance consistent with the classical robustness heritage of LQ designs, though earned here by the soft tuning. Under a longitudinal-only mismatch the LQRy stays nominal and the PID's clamped altitude loop drifts; under wind and gusts both complete the mission with the PID firmer; both tolerate a plausible +10% identification uncertainty. Finally, both controllers are deployed on an STM32 microcontroller through automatic code generation and compared on a lockstep serial hardware-in-the-loop (HIL) bench that is shown to be bit-deterministic and reproduces the SIL verdict, with, for the synchronized PID deployment, a SIL-HIL residual at the quantization floor of the bus, and a sensitivity study ranking the implementation choices that actually affect embedded agreement. The complete toolchain and data are publicly available.

INDEX TERMS Cascade control, code generation, fixed-wing UAV, hardware-in-the-loop simulation, hybrid VTOL aircraft, LQR control, PID control, robustness, STM32.

I. INTRODUCTION

HYBRID vertical take-off and landing (VTOL) aircraft combine the hovering capability of multirotors with the range and efficiency of fixed-wing flight, and have consequently become one of the fastest-growing classes of unmanned aerial vehicles (UAVs) [1], [2]. The price of this ver-

satility is a control problem that changes structure between flight regimes, and the literature increasingly treats each regime as a control problem in its own right [3]. This article focuses on the fixed-wing flight mode of a hybrid VTOL platform — the mode in which the vehicle spends most of its mission — and on the complete path from controller design

to embedded validation; hover and transition are outside its scope.

A. RELATED WORK

Autopilots for small UAVs remain dominated by nested proportional–integral–derivative (PID) loops, a legacy of their simplicity and field-proven robustness [4], and this architecture underpins the open-source flight stacks in widespread use today [5], [6]. Linear–quadratic (LQ) methods are the natural model-based alternative, and comparisons between the two families have accumulated since the seminal indoor quadrotor study of Bouabdallah et al. [7]: on quadrotor platforms [8], [9], on a commercial mini-drone with experimental PID/LQR/MPC benchmarks [10], and, less frequently, on fixed-wing longitudinal dynamics [11].

Hardware-in-the-loop (HIL) simulation — preceded in practice by software-in-the-loop (SIL) verification on the same models — is the established way to close the gap between simulation and flight: the embedded control system is exercised against a simulated plant, so faults in the embedded implementation surface on the bench rather than in the air [12], [13]. The practice extends from classical avionics benches to recent microcontroller-class autopilot platforms [14], [15], is integral to model-based development workflows in which the flight code is generated automatically from the verified model [16]–[18], and has been applied to hybrid VTOL vehicles in the Brazilian research context [19]. Within our research group, this lineage runs through the Pegasus autopilot HIL architecture [20], X-Plane-based HIL of small-UAV attitude control under atmospheric disturbances [21], and the bench-validated fixed-wing autopilot of Santos [22], [23], whose cascade methodology this work adopts. The aircraft models in this line of work are identified from flight-test data [24]–[26].

B. RESEARCH GAPS

Three gaps recur in this comparative literature:

- 1) Platform. Most PID–LQ comparisons target rotary-wing platforms; the fixed-wing mode of a hybrid VTOL vehicle is rarely the object of study.
- 2) Mission-level embedded evaluation. The comparisons are usually confined to simulation of isolated channels, rather than a full mission exercising the longitudinal and lateral–directional loops together on the embedded implementation.
- 3) Robustness. Arguably where the two design philosophies should differ most, robustness is seldom part of the protocol, even though LQ designs carry classical margin guarantees [27] that do not survive output feedback in general [28].

C. CONTRIBUTIONS

The present work addresses these gaps, extending the group lineage in three directions: the plant is the fixed-wing mode of a hybrid VTOL aircraft rather than a conventional fixed-wing one; the embedded controller is not hand-coded but

automatically generated from the same Simulink model that was verified in simulation, so that the tested artifact is the production code itself; and two different control laws — a cascade PID and an output-feedback LQR (LQRy) — are designed against a common, literature-grounded requirements set, stressed under model mismatch and wind, and deployed on the same bench under identical conditions.

The main contributions of this work are:

- the design of a cascade PID architecture for the fixed-wing flight mode of a hybrid VTOL aircraft, evaluated against a fixed-gain output-feedback LQR (LQRy) designed for the same platform, both judged against a common set of literature-grounded closed-loop requirements;
- a robustness study that inverts the nominal verdict: a severe, sign-changing lateral–directional model mismatch makes the aggressively tuned cascade PID lose the mission outright, while the soft fixed-gain LQRy completes it with near-nominal error — with the remaining scenarios adding nuance: under a longitudinal-only mismatch the LQRy stays nominal while the PID’s clamped altitude loop drifts, under wind and gusts the PID is the firmer law, and both tolerate a plausible +10% identification uncertainty;
- a low-cost HIL architecture based on automatic code generation and a lockstep serial protocol, whose SIL–HIL residual is shown to reach the quantization floor of the communication bus for the synchronized PID deployment;
- the characterization of the lockstep HIL loop as a deterministic system — demonstrated by bit-exact reproduction of full 300-second flight cases across two microcontroller models — and its use to quantify how much each implementation choice affects SIL–HIL agreement; and
- a controlled, like-for-like comparison of the two controllers on the same rig, same references, and same metrics, with the known asymmetries of the two design harnesses disclosed explicitly.

D. ORGANIZATION

Section II presents the nonlinear model, the trim condition, and the linearization. Section III details the closed-loop requirements and both controller designs. Section IV reports the SIL simulations and the controller comparison. Section V presents the robustness study. Section VI describes the HIL development, the determinism and sensitivity analyses, and the embedded results. Section VII concludes the article.

II. NONLINEAR MODEL, TRIM, AND LINEARIZATION

This section establishes the plant on which everything else is built: the six-degree-of-freedom nonlinear model of the aircraft with its flight-test-identified aerodynamics, the numerically computed trim condition that anchors every simulation and embedded test of this article, the linearized longitudinal and lateral–directional models used for controller design, and



FIGURE 1. The hybrid VTOL platform used as the target aircraft in this work, developed at the Instituto Tecnológico de Aeronáutica (ITA). The four hover rotors are carried on twin booms and are inactive in the fixed-wing flight mode studied here, in which the aircraft flies on its forward propeller and aerodynamic surfaces. The stability and control derivatives used throughout were identified from flight tests of this airframe [26].

the open-loop stability analysis that motivates closed-loop control — the aircraft carries an unstable spiral mode.

A. NONLINEAR MODEL

The aircraft, shown in Fig. 1, is a small hybrid VTOL platform with a conventional fixed-wing configuration for cruise: four hover rotors are carried on twin booms mounted on the wing, and a separate propeller drives the vehicle in forward flight. Its physical parameters are $m = 2.2$ kg, wing area $S = 0.27$ m², span $b = 1.2$ m, mean aerodynamic chord $\bar{c} = 0.226$ m (aspect ratio $AR = 5.33$, Oswald factor $e = 0.8$), and inertia tensor (kg m²)

$$\mathbf{I} = \begin{bmatrix} 0.1441 & 0 & -0.00167 \\ 0 & 0.1155 & 0 \\ -0.00167 & 0 & 0.2572 \end{bmatrix}. \quad (1)$$

The complete six-degree-of-freedom nonlinear model of this aircraft [29], [30] follows the standard rigid-body formulation [31] and is implemented in Simulink with a state vector of 14 states (body velocities, angular rates, Euler angles, and navigation states). The aerodynamic model uses the usual coefficient build-up,

$$\begin{aligned} C_L &= C_{L_0} + C_{L_\alpha} \alpha + C_{L_q} \frac{q\bar{c}}{2V_0}, & C_D &= C_{D_0} + \frac{C_L^2}{\pi e AR}, \\ C_m &= C_{m_0} + C_{m_\alpha} \alpha + C_{m_{\delta_e}} \delta_e + C_{m_q} \frac{q\bar{c}}{2V_0}, \end{aligned} \quad (2)$$

with analogous expressions for the lateral-directional coefficients C_Y , C_l , and C_n in β , p , r , δ_a , and δ_r . The stability and control derivatives were obtained by system identification from flight-test data of the actual aircraft [26] (longitudinal Model 4 and lateral-directional Model 2 of that identification campaign), following the time-domain output-error practice that is standard for flight-vehicle identification [32] and for small UAVs in particular [33]. Thrust is modeled as $T = T_{max} \delta_t (\rho/\rho_0)^{0.8}$ with $T_{max} = 14$ N.

In the fixed-wing flight mode considered here the hover rotors are inactive, so the plant input vector comprises the

four aerodynamic commands and the three components of the wind velocity, the latter expressed in the local NED frame:

$$\mathbf{U} = \underbrace{[\delta_t \ \delta_e \ \delta_a \ \delta_r]}_{\text{commands}} \mid \underbrace{[w_N \ w_E \ w_D]}_{\text{wind (NED)}}^\top. \quad (3)$$

The wind components are rotated into body axes and subtracted from the inertial velocities before the aerodynamic triad is formed, so that airspeed, angle of attack, and sideslip are computed from the air-relative velocity, $V_T = \|[u \ v \ w]^\top - L_{bn}[w_N \ w_E \ w_D]^\top\|$, in the usual way [31]. Wind therefore enters the model as a physical disturbance input to the aerodynamics rather than as an additive signal on the outputs; it is identically zero at trim and in the nominal simulations, and is exercised as a disturbance in the robustness study of Section V.

Actuator dynamics are included explicitly, since the controllers are ultimately deployed on hardware: each control surface is limited to $\pm 25^\circ$ in position and $150^\circ/\text{s}$ in rate, followed by a first-order lag of $\tau_{act} = 0.05$ s (20 rad/s bandwidth); the throttle channel is limited to $[0, 1]$ with a rate limit of 1 s^{-1} and a spool-up lag of $\tau_{eng} = 0.30$ s. Without these, a reference step produces an unrealizable impulsive surface deflection ($\sim 400^\circ/\text{s}$ in a single step) that no servo — and no HIL target — could reproduce.

B. TRIM CONDITION

The aircraft is trimmed numerically, in the standard formulation of the general aircraft trim problem [31], [34]: the steady, wings-level constraints ($v = p = q = r = \phi = 0$, $u = V_T^e \cos \alpha$, $w = V_T^e \sin \alpha$, $\theta = \alpha + \gamma^e$) are imposed by construction, reducing the equilibrium condition $\dot{x} = f(x, \delta) = 0$ to a square system of three nonlinear algebraic equations — the axial-force, normal-force, and pitching-moment balances $(\dot{u}, \dot{w}, \dot{q}) = 0$, evaluated on the full nonlinear dynamics — in the three unknowns $(\alpha^e, \delta_t^e, \delta_e^e)$. Because the identified aerodynamic build-up makes these equations transcendental, with no closed-form solution, the system is solved with Matlab's `fsolve` (trust-region dogleg, a Newton-type method). The trim sought is steady, wings-level horizontal flight at airspeed $V_T^e = 12.0$ m/s, altitude $h^e = 600$ m, and flight-path angle $\gamma^e = 0$; the resulting equilibrium is stated explicitly below, as it anchors every simulation and every embedded test in this article.

State vector at trim (body axes; unlisted states are zero):

$$\begin{aligned} u^e &= 11.621 \text{ m/s}, & w^e &= 2.992 \text{ m/s}, \\ \theta^e &= 0.25196 \text{ rad} = 14.44^\circ, & h^e &= 600 \text{ m}, \end{aligned} \quad (4)$$

so that $V_T^e = \sqrt{(u^e)^2 + (w^e)^2} = 12.0$ m/s and, since $\alpha^e = \arctan(w^e/u^e) = 14.44^\circ = \theta^e$, the flight-path angle is indeed $\gamma^e = \theta^e - \alpha^e = 0$.

Output vector at trim:

$$V_T^e = 12.0 \text{ m/s}, \quad \alpha^e = 14.44^\circ, \quad \beta^e = 0, \quad \gamma^e = 0, \quad (5)$$

with $p^e = q^e = r^e = 0$, $\phi^e = \psi^e = 0$, $\theta^e = 14.44^\circ$, and $h^e = 600$ m.

TABLE 1. Open-loop modes at the trim condition ($V_T^e = 12$ m/s, $h^e = 600$ m).

Mode	Eigenvalues [s^{-1}]	ω_n [rad/s]	ζ
Short period	$-3.39 \pm 5.31i$	6.30	0.54
Phugoid	$-0.094 \pm 1.11i$	1.12	0.085
Altitude	-1.7×10^{-4}	—	—
Roll subsidence	-11.33	—	—
Dutch roll	$-2.08 \pm 4.59i$	5.04	0.41
Spiral (unstable)	+0.233	—	—
Heading	0	—	—

Command vector at trim (control surfaces and throttle):

$$\delta^e = \begin{bmatrix} \delta_t^e \\ \delta_e^e \\ \delta_a^e \\ \delta_r^e \end{bmatrix} = \begin{bmatrix} 0.2836 \\ 0.1330 \text{ rad } (7.62^\circ) \\ 0 \\ 0 \end{bmatrix}, \quad (6)$$

with the hover-rotor commands identically zero in this flight mode. Throughout the article, the controllers act on increments around this trim; the trim commands of (6) are added at the controller output (see the remark in Section IV-A).

C. LINEARIZATION

The nonlinear model is linearized about the trim condition (4)–(6), decoupling into the standard longitudinal and lateral-directional subsystems used for controller design in Section III. With longitudinal states $x_{lon} = [u \ \alpha \ q \ \theta \ h]^T$ and inputs $[\delta_t \ \delta_e]^T$:

$$A_{lon} = \begin{bmatrix} 0.124 & 13.29 & -2.643 & -9.497 & 0 \\ -0.135 & -2.989 & 0.751 & 0 & 0 \\ 0 & -38.75 & -4.108 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -12 & 0 & 12 & 0 \end{bmatrix}, \quad B_{lon} = \begin{bmatrix} 6.071 & 0 \\ -0.126 & 0 \\ 0 & 45.98 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (7)$$

and with lateral-directional states $x_{lat} = [\beta \ p \ r \ \phi \ \psi]^T$ and inputs $[\delta_a \ \delta_r]^T$:

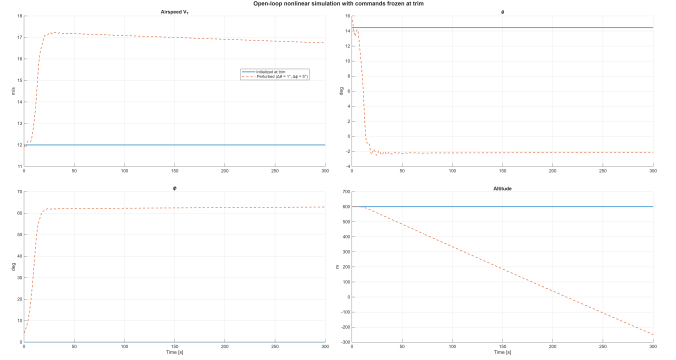
$$A_{lat} = \begin{bmatrix} -0.478 & 0.220 & -0.967 & 0.791 & 0 \\ -51.67 & -12.43 & 9.362 & 0 & 0 \\ 15.74 & -0.648 & -2.355 & 0 & 0 \\ 0 & 1 & 0.257 & 0 & 0 \\ 0 & 0 & 1.033 & 0 & 0 \end{bmatrix}, \quad B_{lat} = \begin{bmatrix} 0 & 0.040 \\ -132.9 & -12.69 \\ 3.282 & 19.09 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (8)$$

(entries below 10^{-3} shown as zero).

D. OPEN-LOOP STABILITY

Table 1 lists the eigenvalues of A_{lon} and A_{lat} and the corresponding classical modes [35], [36]. The longitudinal dynamics are stable, with a well-damped short period and a lightly damped phugoid ($\zeta = 0.085$). The lateral-directional dynamics, however, contain an unstable spiral mode at $s = +0.233 s^{-1}$ — an amplitude doubling time of only $t_2 = \ln 2 / 0.233 \approx 3.0$ s — so the aircraft is open-loop unstable and requires closed-loop stabilization for any practical operation.

This analysis is corroborated by an open-loop simulation of the nonlinear model with the commands frozen at the trim values (6) (Fig. 2). Initialized exactly at the trim state (4), the aircraft remains at equilibrium for the full 300 s (deviations below 10^{-5}), confirming the trim point. Initialized with a

**FIGURE 2.** Open-loop nonlinear simulation with commands frozen at trim. Solid: initialized exactly at trim (remains at equilibrium). Dashed: small initial perturbation — the unstable spiral mode diverges.

small attitude perturbation ($\Delta\theta = 1^\circ$, $\Delta\phi = 5^\circ$) — an offset representative of a minor upset, whose exact size is immaterial since the divergence is exponential — the unstable spiral mode takes over: the bank angle diverges to $\approx 63^\circ$; the aircraft accelerates and enters a steadily descending spiral — the behavior predicted by Table 1. No other disturbance is applied; the perturbation acts only through the initial condition.

III. CONTROLLER DESIGN

This section presents the two control laws under comparison. It first states the closed-loop requirements against which both are judged; then details the cascade PID — stability augmentation, longitudinal and lateral-directional loops, and the tuning of record — summarizes the fixed-gain output-feedback LQR (LQRy) design; and closes by quantifying the roll-to-longitudinal coupling that both designs deliberately leave to feedback.

A. CLOSED-LOOP REQUIREMENTS

Both control laws are judged against the closed-loop requirements of Table 2, adopted from the design practice of our research group [37] and anchored, row by row, in the flight-control literature. The stability-margin rows descend from the military flight-control-system specification MIL-F-9490D, which requires a gain margin of at least 6 dB and a phase margin of at least 45° for the augmented airplane [38] — limits carried into present-day industry practice by its successor SAE AS94900 [39]; the 10 dB adopted here is deliberately conservative. The damping row is stricter than the flying-qualities levels of MIL-F-8785C and MIL-STD-1797A, whose Level-1 bands admit short-period and Dutch-roll damping well below 0.75 [40]–[42]; the $\zeta > 0.75$ target reflects the classical well-damped second-order response ($\zeta \approx 0.7$ yields $\approx 5\%$ overshoot [43]). The time-domain rows (settling time, overshoot, steady-state error) are outer-loop performance targets in the sense of classical autopilot design texts [44], [45], sized for this platform's mission profile. Since these criteria were written for piloted aircraft, their transfer to a small UAV is itself a design decision, supported

TABLE 2. Closed-loop design requirements and their scope.

Requirement	Value	Scope
Settling time	< 60 s	guidance channels (V_T , ψ , h)
Overshoot	$< 10\%$	guidance channels (V_T , ψ , h)
Steady-state error	2–5%	guidance channels (V_T , ψ , h)
Damping ratio	$\zeta > 0.75$	augmented inner modes
Gain margin	> 10 dB	each loop, at design
Phase margin	$> 45^\circ$	each loop, at design

by the literature on scaled flying-qualities requirements for unmanned aircraft [46].

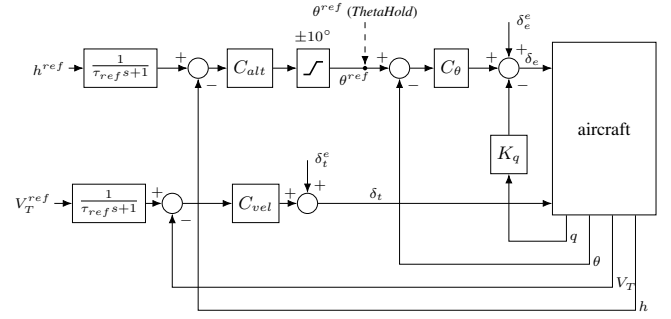
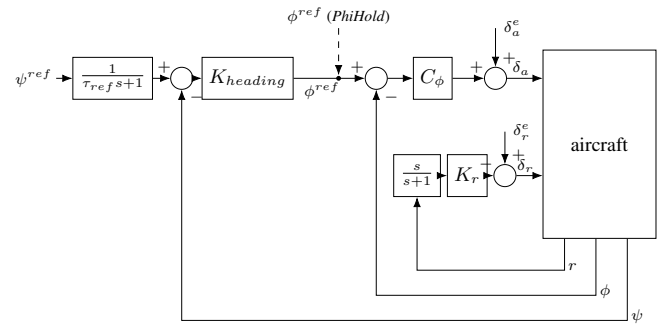
Each row carries an explicit scope. The time-domain rows apply to the guidance-level channels (V_T , ψ , h) on shaped (prefiltered) mission references — they are verified against both controllers in Section IV-E. The damping row applies to the augmented inner modes at design time; the margin rows are enforced per loop during loop shaping (the phase-margin targets of Table 3, 55 – 65° , exceed the requirement). One honest caveat closes the picture: the augmented Dutch-roll damping achieved by the PID design (0.707 , Section III-B) satisfies the MIL-F-8785C Level-1 band with wide margin but sits marginally below the stricter internal target of 0.75 — reported as designed rather than adjusted to fit. On the short-period side, the pitch-rate damper gain of the tuning of record ($K_q = 0.10$, Section III-B) was raised precisely to lift the short-period damping.

B. CASCADE PID ARCHITECTURE

The PID controller follows a classical cascade architecture — successive loop closure in the sense of [47] — designed loop by loop on the linearized models of Section II and then verified on the full nonlinear model. The design methodology — loop ordering, damper-first stability augmentation, and the coordinated-turn heading law — follows [22], developed there for a conventional general-aviation aircraft and applied here to the hybrid VTOL platform in its fixed-wing flight mode.

1) Stability Augmentation

Rate dampers are closed first, tuned by root locus on the linear subsystems. The pitch-rate damper $K_q q$ raises the short-period damping ($K_q = 0.10$ in the tuning of record, a deliberately raised damping target); the yaw damper feeds r through a washout $s/(s+1)$ to the rudder, raising Dutch-roll damping to $\zeta_{DR} \approx 0.707$ without opposing steady turns. The washout yaw damper is one of the oldest devices of automatic flight control: introduced with the first swept-wing jets — which could not be flown unaugmented because of their Dutch roll — in the late 1940s [48], it received its definitive analytical treatment in the classical stability-augmentation literature [44], [49], which the present implementation follows. The roll subsidence mode already meets Level 1 handling qualities with margin ($\lambda = -11.3 \text{ s}^{-1}$, Table 1), so no roll-rate damper is used ($K_p = 0$).

**FIGURE 3.** Longitudinal cascade of the PID law. Dashed: the direct θ^{ref} input of the *ThetaHold* mode.**FIGURE 4.** Lateral-directional cascade of the PID law. Dashed: the direct ϕ^{ref} input of the *PhiHold* mode.

2) Longitudinal Loops

Fig. 3 shows the longitudinal cascade:

- Pitch-attitude inner loop (C_θ): PI on $\theta^{ref} - \theta$ to the elevator, closed over the K_q damper;
- Airspeed hold (C_{vel}): PI on $V_T^{ref} - V_T$ to the throttle;
- Altitude outer loop (C_{alt}): PI on $h^{ref} - h$ to the pitch-attitude reference, with the command clamped to $\pm 10^\circ$ and clamping-type anti-windup. The clamp is an angle-of-attack protection: with θ^{ref} limited to $\pm 10^\circ$ about the trim, the peak α during an altitude capture stays near 16° (the plant model has no stall representation, so the limit is imposed by design).

Depending on the flight mode, the pilot-level selector engages either the altitude cascade (AltitudeHold) or direct pitch tracking (ThetaHold) — the dashed input of Fig. 3 — in which case the altitude loop is open. As the diagram makes explicit, the loops act on increments about trim: every reference enters through the $1/(\tau_{ref}s+1)$ command prefilter detailed below, and the trim commands δ_e^e and δ_t^e of (6) are added at the controller output, downstream of the K_q damper junction. The clamp sits between C_{alt} and the pitch loop, so the anti-windup acts on the altitude integrator exactly when the limit engages.

3) Lateral-Directional Loops

Fig. 4 shows the lateral-directional structure:

- Bank inner loop (C_ϕ): PI on $\phi^{ref} - \phi$ to the aileron (negative gains, since $C_{l_{\delta_a}} < 0$);

TABLE 3. PID loops: design targets and resulting gains (tuning of record).

Loop	Map	K_P	K_I	Original target
C_θ	$\theta^{ref} \rightarrow \delta_e$	0.917	0.857	$\omega_c=2.0$, PM=65°
C_{vel}	$V_T^{ref} \rightarrow \delta_t$	0.32	0.096	$\omega_c=0.8$
C_{alt}	$h^{ref} \rightarrow \theta^{ref}$	0.050	0.0043	$\omega_c=0.35$, PM=55°
C_ϕ	$\phi^{ref} \rightarrow \delta_a$	-0.45	-0.036	$\omega_c=2.0$, PM=65°
$K_{heading}$	$\psi^{ref} \rightarrow \phi^{ref}$	0.1975	—	$\tau_\psi=6$ s
K_q	$q \rightarrow \delta_e$	0.10	—	ζ_{SP} raised
K_r	$r \rightarrow \delta_r$ (washout)	0.1481	—	$\zeta_{DR}=0.707$

- Heading hold ($K_{heading}$): proportional law $\phi^{ref} = K_{heading}(\psi^{ref} - \psi)$ with $K_{heading} = u^e/(g\tau_\psi)$ and $\tau_\psi = 6$ s, the coordinated-turn relation [22]. As with the longitudinal channel, the mode selector engages either heading hold (PsiHold) or direct bank commands (PhiHold) — the dashed input of Fig. 4.

Two structural points are visible in Fig. 4. First, the rudder carries only the washout yaw damper of the stability-augmentation layer — there is no heading- or bank-driven rudder command. Second, the heading reference enters through the same $1/(\tau_{ref}s + 1)$ prefilter as the longitudinal references, and the trim deflections δ_a^e and δ_r^e added at the output are both zero at this flight condition (6).

4) Tuning and Resulting Gains

The compensators are PI ($K_d = 0$): damping is provided by the explicit dampers rather than by derivative action, which also avoids differentiating noisy rate measurements on the embedded target. Each loop was tuned by loop shaping (pidtune) with sequential inner-first closure — every outer loop is designed against the already-closed inner dynamics — targeting the crossover frequencies and margins listed in Table 3, with a bandwidth separation of 5–7× between consecutive loops.

The outer altitude and airspeed loops use two-degree-of-freedom setpoint weighting with $b = 0$ — the proportional term acts on the measured output only, with the reference entering through the integral path — and this tuning of record, retuned against the requirements of Table 2, is the configuration used in every campaign of this article, matching the public repository HEAD.

A command prefilter $1/(\tau_{ref}s + 1)$ with $\tau_{ref} = 3$ s is applied to the increment of each reference before the loops, giving a two-degree-of-freedom structure in the tradition of reference command shaping [50]: the command response is smoothed — the raw step would otherwise demand an unrealizable actuator jump — without altering the feedback design, its margins, or the clamp. This detail matters for the embedded deployment: in the HIL architecture of Section VI the references travel through the link already filtered, so the prefilter stays on the host and only the feedback path is embedded.

With all loops closed, the linear closed-loop system is stable with every pole in the left half-plane: the slowest mode is at -0.127 s⁻¹ and the minimum damping ratio among

the oscillatory modes is $\zeta = 0.55$. Notably, the open-loop unstable spiral mode ($+0.233$ s⁻¹, Table 1) is stabilized at -0.143 s⁻¹.

C. OUTPUT-FEEDBACK LQR (LQRY)

The second control law is a linear-quadratic design with output weighting (LQRY): for each loop, the quadratic cost

$$J = \int_0^\infty (y^\top Q y + u^\top R u) dt, \quad y = C_{syn} x, \quad (9)$$

is minimized over the linearized models of Section II, yielding constant gains on the measured states plus integral action on the tracked error. This is the classical linear-quadratic regulator of Kwakernaak and Sivan [51] in the constant-gain output-feedback setting introduced by Levine and Athans [52] — a problem whose difficulty and solution landscape are surveyed in [53], and whose standard numerical solution is the flight-control-motivated iterative algorithm of Moerder and Calise [54]; the formulation and its application to aircraft control follow the treatment of Stevens et al. [31]. The weight-selection rationale follows established LQ design practice [55], starting from Bryson's rule [56]. The design compared in this article is the fixed-gain LQRY delivered for this platform by the second author and deployed on the embedded bench of Section VI: a single-condition synthesis at the cruise trim of Section II-B (12 m/s, 600 m), deliberately tuned soft. Its synthesis methodology — including the target-transmission-zero augmentation below and the frequency-domain (Nichols) robustness assessment — is detailed in the companion article by the second author [37].

The architecture deliberately mirrors the PID mode structure, so that the two laws face the same tasks: five independent designs — ThetaHold, AltitudeHold (cascading into θ^{ref}), VelHold, PhiHold (a 2×2 design commanding aileron and rudder), and PsiHold (cascading into ϕ^{ref}) — selected by the same three mode switches used by the PID (Table 4). Each synthesis model augments the relevant linear subsystem with a frequency-selective filter

$$G(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{s} \quad (10)$$

on each tracked output — imposing target transmission zeros that shape the closed loop while providing the integral action — plus a first-order actuator lag at 24 rad/s; the constant output-feedback gains are obtained from the coupled Lyapunov equations of the LQ output-feedback problem [52], [54]. The gains are synthesized once, at the cruise trim condition (12 m/s, 600 m), and held fixed over the whole mission — no gain scheduling and no mid-mission switching. The synthesis weights are given in the companion article [37].

The gain set of this fixed-gain delivery — the values exercised in every SIL comparison of Section IV and embedded on the bench of Section VI — is recorded in Table 5 and registered in the repository [30]. Two structural remarks: the pitch design carries the actuator as a fifth synthesis state (the

TABLE 4. LQRy loops: regulated output and actuator (synthesis weights in the companion article [37]).

Loop	Output	Actuator
ThetaHold	θ	elevator
AltitudeHold	$h \rightarrow \theta^{ref}$	elevator
VelHold	V_T	throttle
PhiHold	ϕ, β	ail.+rud.
PsiHold	$\psi \rightarrow \phi^{ref}$	aileron

TABLE 5. Gains of the fixed-gain LQRy delivery (4 significant digits).

Convention: $\Delta u = G_x (x - x^e) + K_I \int e dt$, the increments adding to the trim commands of (6). The K_I entries of PhiHold are the bank (ϕ) integral gains.

Loop	State vector x	G_x	K_I
ThetaHold	$[V_T \alpha q \theta \delta_e]$	$[-0.004694 \ 5.362 \ -10.69 \ -95.34 \ -5.477]$	-25
AltitudeHold	$[V_T \alpha q \theta h]$	$[-0.07210 \ 4.535 \ -0.2142 \ -5.140 \ -0.4439]$	-0.04
VelHold	$[V_T \alpha q \theta]$	$[-0.7506 \ -0.5861 \ 0.006315 \ 0.6642]$	-0.4410
PhiHold (ail.)	$[\beta p r \phi]$	$[0.01697 \ 0.004305 \ 0.01497 \ 0.06072]$	0.003883
PhiHold (rud.)	$[\beta p r \phi]$	$[-0.003758 \ -0.0009542 \ -0.003318 \ -0.01346]$	-0.0008727
PsiHold	$[\beta p r \phi \psi]$	$[-9.147 \ -0.1859 \ 0.1492 \ -1.994 \ -0.6554]$	-0.03366

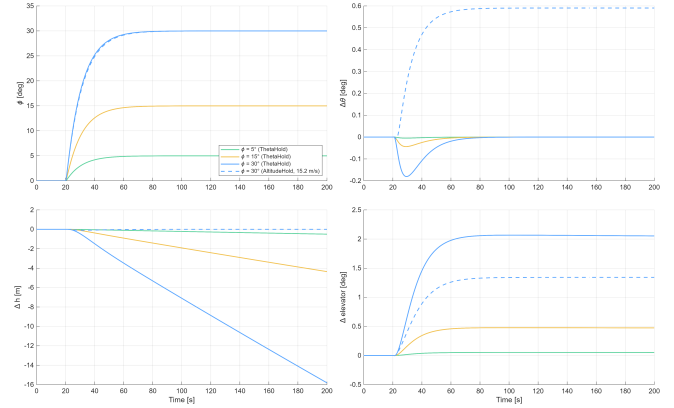
24 rad/s lag), so its gain vector feeds back the elevator alongside the rigid-body states; and the 2×2 lateral design provides for a sideslip-integral gain alongside the bank integral, but in this delivery the β -integral column is numerically zero — only the bank integral is active.

Unlike the PID harness, the LQRy delivery harness contains no actuator models at all: the commanded increments, added to the trim commands of (6), drive the plant inputs directly — no lags, no rate limits, no position or throttle saturations, no reference prefilter, and no θ^{ref} clamp — and references enter the loops as raw steps. The 24 rad/s lag lives in the synthesis models only; on the rig, the fifth gain of the pitch loop multiplies the commanded elevator itself. (Every statement in this paragraph was verified block by block on the archived delivery rig.) These differences between the two design harnesses are consolidated and discussed in Section IV-B.

D. ROLL-TO-LONGITUDINAL COUPLING

Both designs treat the longitudinal and lateral-directional channels independently, so the classical roll-to-pitch coupling — banking tilts the lift vector, and the lift deficit, proportional to $1 - \cos \phi$, drops the nose and costs altitude — is handled by feedback alone. Since this is a deliberate design choice, it was quantified rather than assumed benign on the cascade PID — the law whose clamped altitude loop the robustness study of Section V stresses hardest: bank captures of increasing amplitude (ϕ^{ref} steps of $5/15/30^\circ$ at $t = 20$ s, shaped by the reference prefilter as in every PID run), in ThetaHold (altitude loop open, the worst case) and in AltitudeHold, with all longitudinal metrics taken as deviations from the $\phi = 0$ baseline (Fig. 5).

The results confirm the mechanism and bound its size. In ThetaHold, the transient pitch dip scales with $1 - \cos \phi$ as expected — -0.005° , -0.04° , and -0.18° for $5/15/30^\circ$ — negligible at the 5° amplitude of the flight campaign. With the altitude loop open the altitude drifts by construction —

**FIGURE 5.** Roll-to-longitudinal coupling (PID), amplitude sweep: ϕ^{ref} steps of $5/15/30^\circ$ at $t = 20$ s, prefiltered reference; ϕ , $\Delta\theta$, Δh , and $\Delta\text{elevator}$ relative to the $\phi = 0$ baseline. Solid: ThetaHold at the 12 m/s trim; dashed: 30° in AltitudeHold at 15.2 m/s.

—15.8 m over 200 s at 30° ; closing the altitude loop absorbs the coupling, with θ^{ref} settling higher ($+0.59^\circ$) and Δh returning to zero after a transient dip of only 0.10 m, at the cost of a small steady elevator increment. The commands stay smooth throughout: aileron peak of 1.0° at $1.0^\circ/\text{s}$, courtesy of the same command shaping applied to every reference.

In the steady turn, the elevator increment is larger in the ThetaHold run at the 12 m/s trim (solid trace, 2.06°) than in the AltitudeHold run at 15.2 m/s (dashed, 1.34°), which at first seems backwards — why would the altitude-holding configuration need less elevator? The explanation is the operating point, not the mode: sustaining the same 30° bank (load factor $1/\cos 30^\circ = 1.15$) requires an increment of lift that, at 15.2 m/s, is produced with less angle of attack — and less elevator — because the dynamic pressure is $(15.2/12)^2 = 1.6$ times higher. The measured increment ratio (0.65) sits within a few percent of the dynamic-pressure prediction $(12/15.2)^2 = 0.62$, the residual owing to the θ^{ref} adjustment and the angle-of-attack dependence of drag.

In answer to the natural question — no, no feedforward decoupling term is used by either law ($K_{ff} = 0$ throughout): with the altitude loop closed the residual coupling is 0.10 m even at a 30° bank, and at the 5° amplitude of the flight campaign the $1 - \cos \phi$ factor is a further $35\times$ smaller, so feedback suffices. A feedforward stage, and its evaluation on both laws at larger bank angles, is left as future work (Section VII).

IV. CLOSED-LOOP SIMULATIONS (SIL)

This section evaluates both controllers on the nonlinear model. It opens with two framing remarks — the incremental-command convention about trim, and the comparison protocol with its known harness asymmetries — then validates the PID against its linear design predictions, characterizes the LQRy in its own harness, and closes with the head-to-head comparison on the combined mission, judged against the requirements of Section III.

A. REMARK ON TRIM COMMANDS

All controllers in this work operate on incremental commands about the trim condition of Section II-B: the compensators output $\Delta\delta$, and the trim commands δ^e of (6) are added before the actuator models. Consequently, whenever a simulation reports a command signal, the reader should distinguish the absolute command $\delta = \delta^e + \Delta\delta$ from the increment $\Delta\delta$ produced by the controller. In steady flight at the trim point, the absolute commands settle at δ^e (e.g., the elevator at 7.62°), not at zero.

B. COMPARISON PROTOCOL AND KNOWN ASYMMETRIES

Every head-to-head result in this article follows the same protocol — in the spirit of the experimental controller benchmarks of the recent literature [9], [10]: the same nonlinear plant, the same reference amplitudes and timings, and the same metric windows, with each control law running in its own design harness, exactly as delivered. That last clause matters, because the two harnesses are not identical, and an honest comparison must say so. Three asymmetries are known and uncontrolled:

- 1) Actuator models. The PID harness includes the full actuator chain of Section II (position and rate saturations, 20 rad/s servo and 0.30 s engine lags). The LQRy harness carries no actuator model at all — its commands drive the plant inputs directly; the 24 rad/s lag exists only inside its pitch synthesis model (Section III-C).
- 2) Reference shaping. The PID applies the $1/(3s + 1)$ prefilter of Section III-B to every reference increment; the LQRy receives raw reference steps.
- 3) Protection logic. The $\pm 10^\circ$ θ^{ref} clamp and its anti-windup exist only in the PID cascade.

To bound the effect of these asymmetries on the conclusions, the PID scenarios were repeated in an equalized configuration — rate limits removed, clamp and prefilter disabled, and the actuator lags lightened to 24 rad/s (surfaces) and 0.1 s (throttle) — approaching the LQRy's unfiltered, rigid-command harness. Fig. 6 overlays the two PID configurations on the combined scenario: the equalization shifts the guidance captures moderately — airspeed rise time from 9.5 to 5.8 s and settling from 18.0 to 11.4 s, heading settling from 33.0 to 29.3 s, altitude settling from 44.4 to 40.8 s, all with zero overshoot — without changing any comparative conclusion drawn below.

Two further implementation notes complete the disclosure. First, the original delivery of the fixed-gain LQRy fed back the absolute state instead of the deviation from trim, producing a start-up kick (initial elevator command of 4.11° instead of the trim 7.62°); the corrected delivery — the build compared here — feeds back trim deviations, and its runs were verified to start at the trim command. The embedded (incremental-form) implementation of Section VI was immune to this distinction throughout. Second, the LQRy rig initializes the plant at $V_T = 12.0$ m/s while

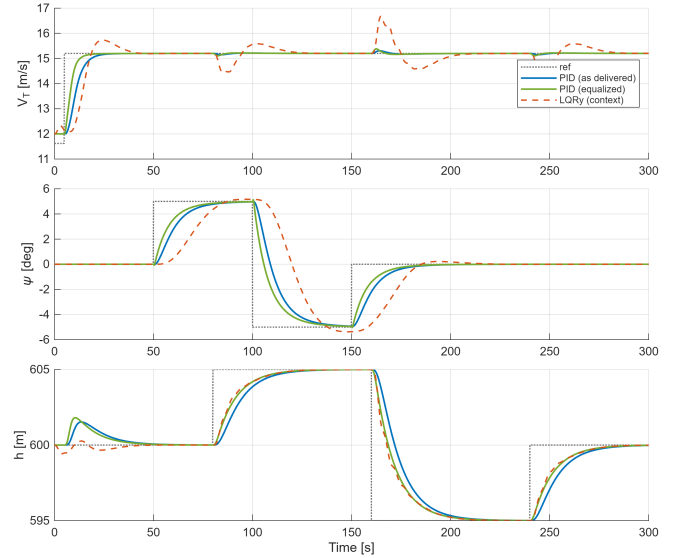


FIGURE 6. Equalization ablation, combined scenario: PID as delivered vs. PID with the harness equalized toward the LQRy rig (24 rad/s actuators, no rate limits, no clamp, no prefilter), with the fixed-gain LQRy response for context.

its stored controller trim corresponds to 11.62 m/s, so LQRy runs begin with a small longitudinal offset that the airspeed loop resolves before the first reference event.

C. PID ON THE NONLINEAR MODEL

The cascade PID was evaluated on the nonlinear model under three maneuvers, with the linear closed-loop response overlaid in each case: a longitudinal climb ($h + 10$ m), a lateral-directional heading change ($\psi + 15^\circ$), and an aggressive combined maneuver ($h + 30$ m and $\psi + 40^\circ$ commanded simultaneously, saturating the θ^{ref} clamp for ≈ 10 s). Fig. 7 shows the combined case. Linear and nonlinear responses agree closely even in the aggressive maneuver (maximum deviations of 0.38 m in altitude, 0.52° in pitch, and 0.27° in bank; below 0.21 m/ 0.75° / 0.05° in the single-axis maneuvers), validating both the linearization and the linear-design-nonlinear-verification workflow. This validation was performed with the original loop-shaping tuning; its conclusion — the fidelity of the linearization — is independent of the subsequent requirements-driven retune.

D. LQRy ON THE NONLINEAR MODEL

The LQRy was evaluated directly on the nonlinear model, in its own harness, on the same mission used for the comparison below. Its closed-loop character is the deliberate opposite of the retuned PID's: soft, low-bandwidth captures — airspeed settling in ≈ 34 s with a visible 16.5% overshoot, heading in ≈ 50 s, altitude in ≈ 41 s (Table 7) — with smooth actuator activity throughout. Unlike the PID case, no linear-overlay validation is presented: the linear design models of Section III-C are the basis of the synthesis, and the nonlinear evaluation here is the verification step of record for the delivered design.

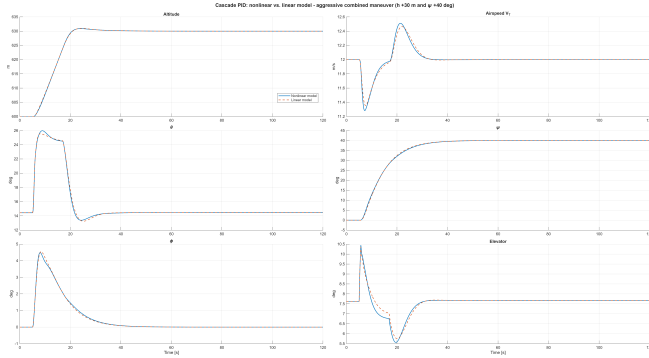


FIGURE 7. PID, aggressive combined maneuver ($h + 30$ m and $\psi + 40^\circ$ simultaneous): nonlinear model vs. linear closed loop.

E. CONTROLLER COMPARISON

The evaluation maneuvers deserve a word of justification. Doublets are the conventional, well-characterized excitation of flight-test practice — originally for identification, where their frequency content is matched to the modes of interest [57], [58] — and step and doublet captures underlie the demonstration requirements of the flying-qualities specifications [40]. For unmanned aircraft, however, the literature argues that mission-based evaluation is the most transferable criterion, since manned handling-qualities metrics do not carry over directly [59]; command-profile tracking of airspeed, altitude, and attitude remains the standard validation practice for small-UAV control laws [60]. The campaign therefore uses both: per-axis guidance scenarios to characterize each channel, and a 300 s combined mission (airspeed step, heading doublet, altitude doublet, overlapping in time) as the mission-level verdict.

The two controllers are compared on the nonlinear model on the 300 s combined mission, with identical reference profiles injected into each rig's unmodified reference harness (VelHold+PsiHold+AltitudeHold engaged):

- 1) Airspeed: step $12 \rightarrow 15.2$ m/s at 5 s;
- 2) Heading: doublet of $\pm 5^\circ$ at 50/100/150 s;
- 3) Altitude: doublet of ± 5 m at 80/160/240 s,

all three overlapping in time (flight case 4 of the campaign of Section VI); the reference events are spaced widely enough that each channel's capture can be read individually. Table 6 reports the mission tracking errors, Table 7 the per-channel metrics against the requirements, and Fig. 8 the time histories. The verdict at guidance level is one-sided in performance terms: the requirements-retuned PID is both the faster and the tighter law — zero overshoot on every channel, captures in 18–45 s, and RMS tracking errors of 0.07 m/s, 1.27° , and 1.65 m — while the soft fixed-gain LQRy concedes a factor of ≈ 5 in airspeed RMS (0.38 m/s) and ≈ 2.3 in heading RMS (2.92°), with the altitude channel essentially matched (1.78 versus 1.65 m). The LQRy's captures are slower on airspeed and heading (34 and ≈ 50 s) and marginally faster on altitude (40.6 versus 45.1 s), always with smooth actuator activity. What the LQRy buys with this softness is not visible

TABLE 6. Combined-mission tracking errors (SIL, identical references, $t > 12$ s).

Channel	PID		LQRy	
	RMS	max	RMS	max
V_T [m/s]	0.07	0.80	0.38	2.35
ψ [$^\circ$]	1.27	5.79	2.92	10.16
h [m]	1.65	6.82	1.78	10.00

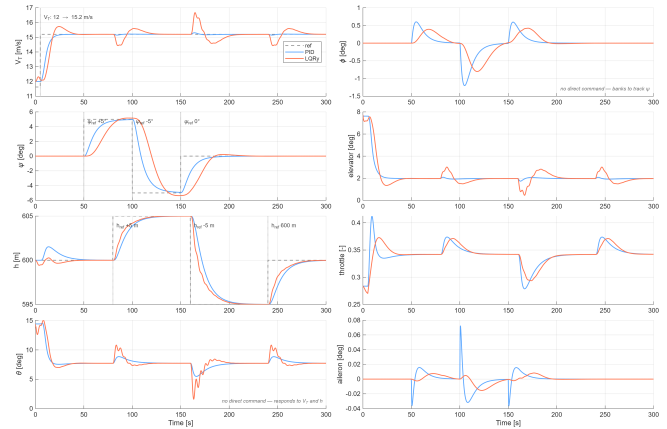


FIGURE 8. Combined mission: PID vs. LQRy on the nonlinear model, identical references — states on the left, actuators on the right.

TABLE 7. Requirements compliance on the combined mission (SIL), guidance channels, identical references.

Requirement	PID			LQRy		
	V_T	ψ	h	V_T	ψ	h
OS < 10% [%]	0.0	0.0	0.0	16.5	3.4	0.0
t_s < 60 s [s]	18.0	38.1	45.1	34.0	≈ 50	40.6
$e_{ss} \leq 5\%$ [%]	0.1	1.1	0.1	0.1	3.1	0.1

on the nominal plant; it is precisely what the robustness study of Section V measures.

Table 7 closes the loop with the requirements of Section III-A, evaluated per channel on the mission's reference events. The PID meets every row. The LQRy meets the settling row on every channel (34–50 s, inside the 60 s limit) and the steady-state row (at most 3.1%), but violates the overshoot requirement on the airspeed channel (16.5% against the 10% line).

The embedded (HIL) campaign of Section VI deploys exactly the two laws compared here — the PID tuning of record and this fixed-gain LQRy — so the bench results of Section VI-G close the same comparison on hardware.

V. ROBUSTNESS STUDY

The comparison of Section IV-E holds for the nominal plant — the very model both controllers were designed on — and on that plant the PID wins on every performance metric. This section asks how much of that picture survives when the plant is no longer the design model. The protocol is deliberately strict: only the plant changes. Both controllers keep the gains and trim commands of the nominal design, and every run

repeats the combined mission of Section IV-E (V_T 12 $\rightarrow$ 15.2 m/s at 5 s, heading doublet $\pm 5^\circ$ at 50/100/150 s, altitude doublet ± 5 m at 80/160/240 s, 300 s total) with the same reference set used throughout Section IV. Four scenarios are considered, spanning plausible-to-severe model uncertainty and an exogenous disturbance: (A) a severe model mismatch, replacing the complete aerodynamic set by an independent identification of the same airframe; (B) the longitudinal half of that mismatch only, to isolate the origin of the failure observed in (A); (C) a plausible identification uncertainty, multiplying all 23 stability and control derivatives by 1.10; and (D) wind and gusts applied through the native disturbance inputs of (3). Table 8 summarizes the mission tracking errors — the degradation columns normalize each scenario by the same controller’s nominal performance, so that the two laws are compared against themselves rather than against each other — and Table 9 tracks the LQRy’s requirements scorecard across the scenarios.

A. SEVERE MODEL MISMATCH

The perturbed plant replaces the nominal, flight-test-identified aerodynamic set [26] by the complete alternative set originally distributed with the nonlinear model implementation [29], [30], obtained through an independent identification of the platform. The two sets differ substantially — the pitch-damping derivative C_{m_q} by a factor of 4, the elevator effectiveness $C_{m_{\delta_e}}$ by 3.3, and the lateral-directional set even changes the sign of C_{y_β} and C_{y_p} — so this scenario probes well beyond parametric uncertainty: it is a structurally different aerodynamic description.

This is the scenario that inverts the nominal verdict. The PID — the faster, tighter law of Section IV-E — loses the mission: the heading doublet triggers a violent bounded oscillation, the heading error grows to $\sim 2100^\circ$ (the aircraft flies repeated full circles), pitch swings of tens of degrees appear, and the altitude sinks by hundreds of meters (Table 8: RMS heading error of 819° , altitude error of 554 m). The fixed-gain LQRy, on the very same perturbed plant, completes the mission with near-nominal error: degradation factors of 0.93–0.99 on every guidance channel (Fig. 9), and a requirements scorecard essentially unchanged from the nominal one (Table 9). A law that concedes a factor of 5 in nominal airspeed RMS absorbs, without retuning, a plant whose lateral derivatives changed sign — the robustness heritage classically associated with LQ designs [27].

That heritage, however, is not a theorem here: the guaranteed margins of LQ state feedback do not survive output feedback in general [28], so the tolerance observed is an empirical property of this delivery — earned by its deliberately soft tuning — rather than a structural guarantee.

B. ISOLATING THE ORIGIN: LONGITUDINAL MISMATCH ONLY

Scenario B replaces only the longitudinal derivatives by the mismatched set, keeping the nominal lateral-directional model. Both laws now complete the mission, which identifies

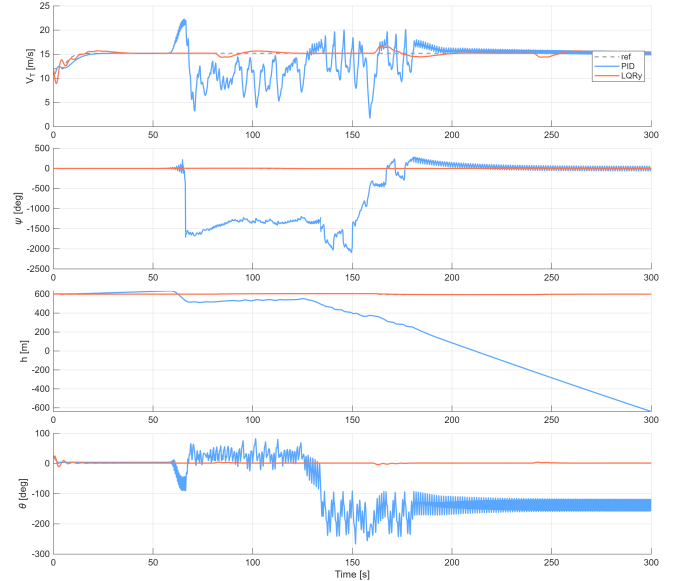


FIGURE 9. Scenario A (severe mismatch): combined mission on the perturbed plant, nominal-design controllers. The PID enters a violent bounded oscillation and loses the mission; the fixed-gain LQRy completes it with near-nominal error.

the lateral-directional half of the mismatch — with its sign inversions — as what defeats the PID in Scenario A. The LQRy again stays near nominal in every channel (Table 9). The PID maintains heading and airspeed at nominal levels (degradation of 1.01 and 1.14), but the longitudinal mismatch alone is not benign for it: the altitude channel degrades by a factor of 69 (RMS 113.8 m, drift of +193 m), a steady drift of the altitude hold. The mechanism is specific: the θ^{ref} clamp of $\pm 10^\circ$ (Section III-B), sized for the control authority of the nominal model, leaves the altitude loop without enough authority on the perturbed plant, and the loop drifts without ever diverging.

C. PLAUSIBLE IDENTIFICATION UNCERTAINTY

Scenario C multiplies all 23 stability and control derivatives by 1.10 — a uniform +10% multiplicative perturbation with preserved signs, representative of a realistic identification error, in the spirit of the perturbed-derivative robustness campaigns of the fixed-wing UAV literature [61]. Both controllers are insensitive to it: every PID degradation factor lies between 1.00 and 1.03, the LQRy scorecard is essentially the nominal one (Table 9), and the responses are visually indistinguishable from the nominal runs. Within plausible identification uncertainty, both designs are robust; the contrast between them only appears under the severe, sign-changing mismatch of Scenarios A–B.

D. WIND AND GUSTS

Small, low-speed fixed-wing UAVs are disproportionately sensitive to atmospheric disturbance — gust velocities are a large fraction of their airspeed [62] — so no comparison of flight controllers for this class is complete without a wind

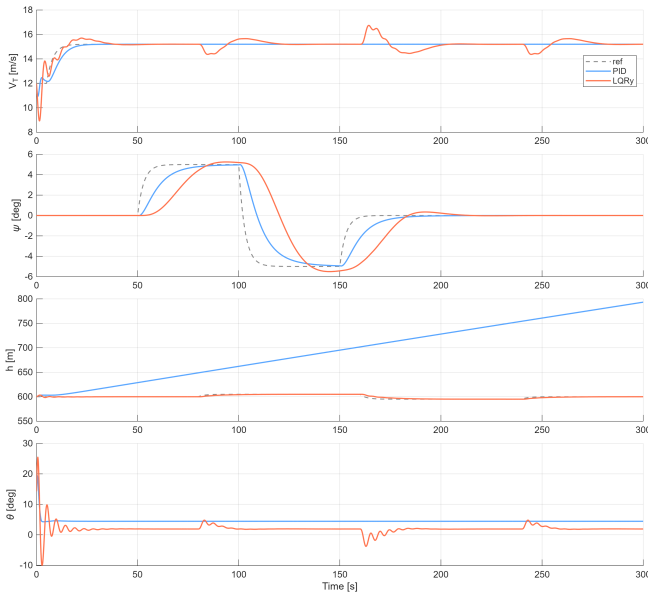
TABLE 8. Robustness campaign: RMS tracking errors ($t > 12$ s) and degradation relative to the nominal scenario (RMS scenario / RMS nominal). Gains and trim always from the nominal design.

Scenario	Ctrl.	RMS error			Degradation ($\times$ nominal)		
		V_T [m/s]	ψ [$^\circ$]	h [m]	V_T	ψ	h
Nominal	PID	0.071	1.27	1.65	—	—	—
	LQRy	0.38	2.92	1.78	—	—	—
(A) Severe mismatch	PID	loses the mission ^a					
	LQRy	0.353	2.61	1.77	0.93	0.89	0.99
(B) Longitudinal only	PID	0.081	1.28	113.8	1.14	1.01	68.9
	LQRy	0.353	2.94	1.77	0.93	1.01	0.99
(C) All derivatives $\times 1.10$	PID	0.073	1.28	1.65	1.03	1.01	1.00
	LQRy	0.375	2.93	1.77	0.99	1.00	1.00
(D) Wind and gusts	PID	0.104	2.24	2.55	1.46	1.76	1.54
	LQRy	0.481	3.68	5.21	1.26	1.26	2.93

^aViolent bounded oscillation triggered by the heading doublet: RMS errors of 3.14 m/s, 818.7 $^\circ$, and 554.0 m ($44\times/644\times/335\times$ nominal); see Section V-A.

TABLE 9. LQRy requirements scorecard across the robustness scenarios (mission reference events, $V_T/\psi/h$). The nominal row repeats Table 7.

Scenario	OS [%]	t_s [s]	e_{ss} [%]
Nominal	16.5 / 3.4 / 0	34 / ≈ 50 / 41	0.1 / 3.1 / 0.1
(A) Severe mismatch	15.8 / 0 / 0	31 / ≈ 50 / 40	1.1 / 6.5 / 0.1
(B) Longitudinal only	15.8 / 5.1 / 0	31 / ≈ 50 / 40	1.1 / 4.7 / 0.2
(C) $\times 1.10$	12.5 / 3.8 / 0	35 / ≈ 50 / 41	0.9 / 3.5 / 0.1
(D) Wind and gusts	42.8 / 72.8 / 0	45 / ≈ 50 / >80	15.1 / 15.2 / $>>$

**FIGURE 10.** Scenario B (longitudinal mismatch only): the PID's altitude hold drifts +193 m while its heading and airspeed stay nominal; the LQRy remains near nominal on every channel.

scenario. Scenario D perturbs the nominal plant through the wind inputs of (3): a 4 m/s tailwind ramped in over 20–25 s, two lateral $1 - \cos$ gusts of ± 3 m/s (55–65 s and 105–115 s, timed over the heading doublet), and two vertical $1 - \cos$ gusts of ± 2 m/s (85–95 s and 165–175 s, timed over the altitude doublet). The $1 - \cos$ profile is the classical discrete-

gust idealization [63], retained to this day in the gust conditions of certification regulations [64]; deterministic discrete gusts (rather than stochastic Dryden or von Kármán turbulence [40], [65]) keep the scenario exactly reproducible run to run, consistent with the deterministic-bench philosophy of Section VI. Both controllers remain stable and complete the mission with moderate peak errors — and this is the one disturbance class where the nominal ranking persists rather than inverts. The PID degrades most in heading under the lateral gusts (RMS $\times 1.76$, peak error 12.9 $^\circ$) but stays firm elsewhere. The LQRy feels the wind more on both of its weaker channels: heading peaks of 18.6 $^\circ$ and an altitude degradation of $2.9\times$ its nominal RMS (≈ 5.2 m, peak 11.2 m), with the gust transients sweeping its scorecard readings up accordingly (overshoot entries of 42.8%/72.8% on V_T/ψ in Table 9 are gust excursions measured against the same reference events, not capture overshoots). Neither controller approaches loss of control.

E. DISCUSSION

The four scenarios, read together, draw the central trade-off of this article. On the nominal plant the aggressively tuned cascade PID dominates every performance metric (Section IV-E); under the severe, sign-changing mismatch it is the PID that loses the mission — its fixed loop shapes invalidated by the inverted lateral derivatives — while the soft fixed-gain LQRy absorbs the same plant change with degradation factors below unity on two channels. The longitudinal-only mismatch adds a subtler failure mode on the PID side: an outer loop silently starved by a clamp sized for the nominal plant, drifting without diverging. Under wind the ranking reverts to the nominal one — the PID firmer on every channel, the LQRy conceding altitude and heading peaks — and within plausible identification uncertainty both laws are robust. Robustness to severe mismatch is not a free property of the LQ structure — the classical margins of LQ state feedback [27] are lost under output feedback in general [28] — so the tolerance observed here is an empirical property

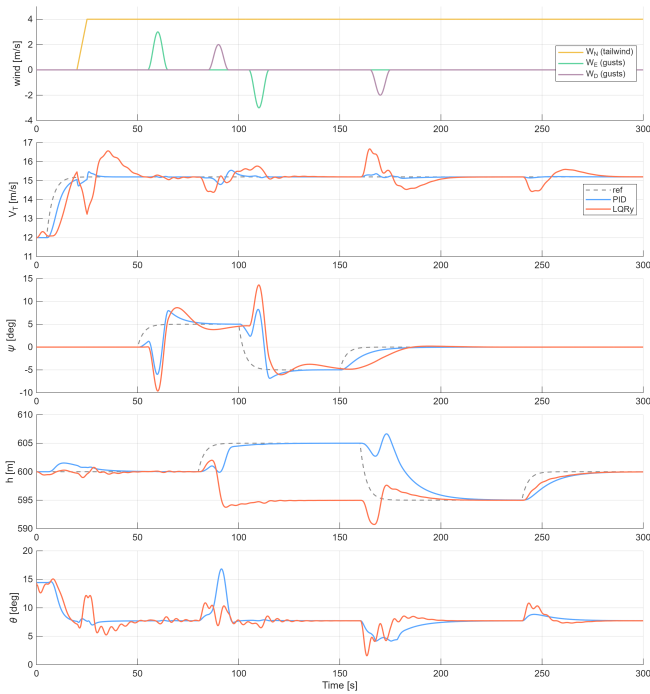


FIGURE 11. Scenario D (wind and gusts): wind profile (NED components) and closed-loop responses.

of this deliberately soft delivery. What the comparison isolates is therefore a genuine performance–robustness trade-off between the two standard laws as delivered: the cascade PID converts its bandwidth into nominal tracking quality; the fixed-gain LQRy converts its softness into tolerance of model error. Robust-control syntheses that shape this trade-off explicitly have been pursued for small fixed-wing UAVs in our own research context [66]; here the point is narrower: which of the two standard laws is preferable depends on whether the dominant threat is model error — where the PID’s failure is catastrophic and the LQRy’s tolerance is decisive — or atmospheric disturbance, where the PID holds the line more tightly.

VI. HARDWARE-IN-THE-LOOP DEVELOPMENT

HIL simulation is the established bridge between simulation and flight for UAV autopilots [12], [13], and the bench lineage of our group runs from the Pegasus real-time HIL architecture [20] through X-Plane-based attitude HIL [21] to the fixed-wing autopilot bench of Santos [22]. This section describes the present incarnation: a lockstep serial bench in which the embedded controller is the auto-generated production code. It covers the controller discretization and code generation, the bench architecture and its link-level validation, the bit-determinism of the lockstep loop and the sensitivity study that determinism enables, and the embedded campaign results — closing with the controller comparison repeated on the hardware.

A. CONTROLLER DISCRETIZATION

Both controllers are discretized at $T_s = 10$ ms directly in Simulink (automatic conversion of the compensators to their discrete-time forms; the washout filter via Tustin transformation), producing single-rate discrete models suitable for code generation. The discretized PID was verified against the continuous design on the nonlinear model, with an RMS deviation below 0.1% of each command amplitude — discretization is not a significant error source at this sample rate, as quantified in Section VI-E.

A practical note on output saturation: applying the automatic conversion to the complete control loop with the output saturation blocks included did not yield a working discrete model. The procedure adopted was to discretize the control law alone and to re-insert the saturations after discretization. In the deployment reported here the saturation limits (throttle in $[0, 1]$, surfaces at $\pm 25^\circ$) are enforced in the embedded C code, mirroring the host-side limits exactly.

B. HIL ARCHITECTURE

Fig. 12 shows the architecture. The discrete controller is translated to C by Simulink Embedded Coder (`ert.tlc`, ARM Cortex-M, double precision) — the automatic-code-generation workflow of model-based aerospace software practice [16], [17] — and embedded in an STM32 microcontroller running FreeRTOS — a class of target for which HIL validation remains an active topic [15]. The host PC runs only the nonlinear plant, the reference generation, and the serial interface. Host and target exchange one frame per controller period over a 460 800-baud serial link: 16 single-precision floats to the target (states, filtered references, and mode switches) and 4 floats back (throttle, elevator, aileron, rudder), delimited by a fixed header (`0xDE 0xAD`) and terminator.

The link operates in lockstep, the mechanism by which simulator and autopilot advance strictly by exchanged messages to make runs deterministic and reproducible — standard practice in simulation stacks such as PX4 [67]: the plant blocks at each 10 ms simulation step until the controller reply arrives, so simulated time advances only upon a completed exchange. Every HIL sample therefore corresponds one-to-one to a SIL sample — the comparison is free of timing jitter by construction.

Two aspects of the campaign design are worth stating explicitly, as they are what make the SIL and HIL results directly comparable:

- Single source of references. One script generates both the SIL and the HIL runs of every flight case, so the two campaigns cannot drift apart in their reference definitions.
- Mode selection over the link. The three flight-mode switches travel in the frame rather than being compiled into the firmware, so a single binary serves all flight cases of the PID deployment and the target is never reprogrammed mid-campaign (the LQRy deployment deviates from this, as noted in Section VI-F).

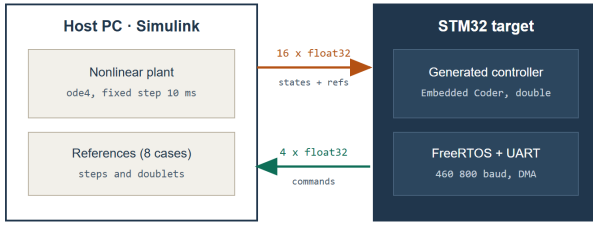


FIGURE 12. HIL loop in lockstep: nonlinear plant on the host, generated controller code on the STM32, binary serial frames at each 10 ms step.

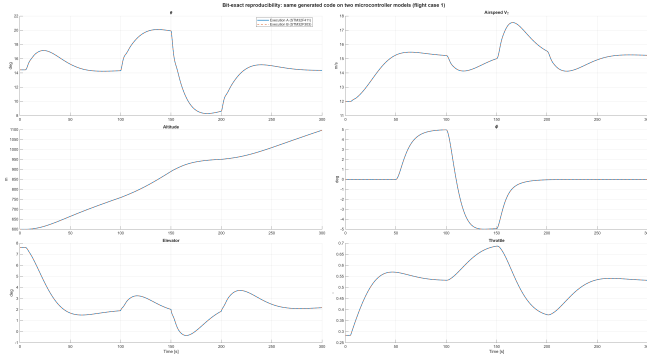


FIGURE 13. Bit-exact reproducibility: the same generated controller code executed on two different microcontroller models over a 300-second flight case. Maximum difference across all channels is identically zero.

C. LINK AND CHAIN VALIDATION

Before any campaign, each element of the chain was validated independently: (i) serial echo at 460 800 baud with zero errors; (ii) a frame-level test in which the embedded controller response to 200 known input frames matched the simulated discrete controller to within 4×10^{-8} (the single-precision quantization of the bus); and (iii) the discretization check of Section VI-A.

D. DETERMINISM OF THE LOCKSTEP LOOP

A lockstep HIL loop built from double-precision arithmetic on both sides, with single-precision quantization only on the wire, has no source of run-to-run variability: there is no timing race, no dropped-sample interpolation, and no floating-point reduction whose order can change. The loop should therefore be bit-deterministic.

We verified this property directly. The generated LQR controller was compiled and executed on a microcontroller model different from the one used in an earlier execution of the same code (STM32F303 versus STM32F411, different core configurations and clock trees), driving the same host model. Over a full 300-second flight case, the two executions produced bit-identical outputs on all recorded channels (maximum difference identically zero, Fig. 13); repeating additional flight cases of the campaign reproduced the same result.

This is a useful property for a validation bench, and not merely a curiosity. Because the loop contributes no noise of its own, any deviation observed between two runs is

TABLE 10. Sensitivity of SIL–HIL agreement: each factor injected individually into the validated bench (RMS vs. the clean SIL run, flight case 1).

Factor	θ [°]	h [m]
Baseline (unmodified bench)	0.030	0.09
S1: start-up desynchronization (7 s)	0.014	0.07
S2: airspeed reference mismatch (3 m/s)	0.447	258
S3: controller discretization	0.003	0.001
S4: 1% frame loss	0.032	1.30

attributable to a deliberate change in the setup, which makes the bench a measuring instrument rather than a source of scatter. It also means that a campaign can be reproduced exactly by a third party, on different hardware, from the archived code and reference definitions — a stronger form of reproducibility than the usual statistical agreement. The sensitivity study of the next subsection relies on this: each experiment changes exactly one factor, and the resulting difference is entirely due to that factor.

This determinism also settles a practical concern of embedded deployment: block execution order in a graphical model can, in principle, alter the generated real-time code with respect to the simulation. Here, with both controllers auto-generated from single-rate discrete models, the cross-MCU bit-exactness rules out target-side nondeterminism, and the quantization-floor SIL–HIL residual of the synchronized PID deployment (Section VI-F) bounds any execution-order effect at the level of the bus quantization itself.

E. SENSITIVITY OF SIL–HIL AGREEMENT

Deploying a control law on an embedded target involves a number of implementation choices whose effect on SIL–HIL agreement is rarely quantified. Exploiting the determinism established above, we injected each factor individually into the validated PID bench and measured the resulting discrepancy against the clean SIL run (Table 10). All experiments use flight case 1; the baseline row is the unmodified bench.

The ranking is informative, and in part counter-intuitive:

1) Reference consistency dominates (S2)

An airspeed reference differing by 3 m/s between the SIL and HIL campaigns — a plausible outcome whenever the two campaigns are configured from separate scripts — produces a 258 m RMS altitude discrepancy. This is not a fidelity loss of the HIL chain at all: it is a comparison between two different flight conditions. It is, by two orders of magnitude, the largest effect measured, which motivates the single-source-of-references design of Section VI-B.

2) Start-up desynchronization leaves a persistent residual (S1)

Embedded controllers commonly hold their integrators in reset until the communication link engages, so the target starts from the trim command while the simulated controller starts from its initial states and produces an initial transient. The two trajectories then differ over the first seconds and,

TABLE 11. PID: RMS SIL–HIL discrepancy per flight case ($t > 12$ s).

Case	θ [°]	V_T [m/s]	h [m]	ϕ [°]
1	0.0028	0.0009	0.089	$\sim 10^{-6}$
2	0.0007	0.0002	0.004	$\sim 10^{-6}$
3	0.0035	0.0011	2.75	$\sim 10^{-6}$
4	0.0007	0.0002	0.004	$\sim 10^{-7}$
5	0.0109	0.0059	0.402	$\sim 10^{-6}$
6–8	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-6}$

in modes where an outer loop is open, that difference is integrated and never recovered. Synchronizing the two start-ups — as done in the PID deployment — removes this term; leaving them unsynchronized leaves a residual that no amount of link quality can remove.

3) Frame loss is benign at low rates (S4)

Discarding 1% of the controller replies at random — each discarded reply is replaced by holding the previous command for that step, so the lockstep timing is preserved — degrades altitude agreement to 1.30 m but leaves attitude essentially unaffected, since the actuator rate limits and the airframe dynamics filter isolated command glitches. This is reassuring for links less reliable than the one used here.

4) Discretization is negligible (S3)

The usual first suspect — the continuous-to-discrete conversion of the control law — accounts for 0.003° in attitude and 10^{-3} m in altitude at $T_s = 10$ ms, three orders of magnitude below the reference-consistency term.

F. EMBEDDED CAMPAIGN RESULTS

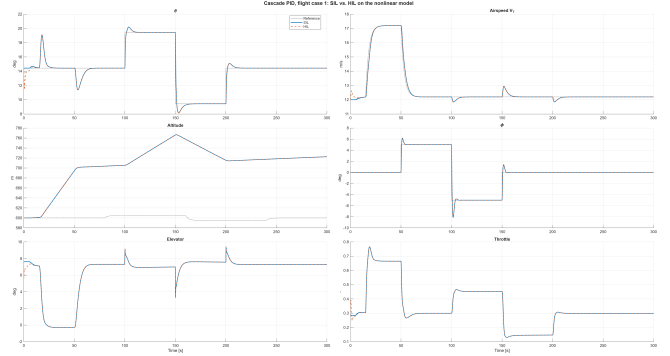
Eight flight cases (all combinations of the three mode switches, with doublets on attitude/heading and steps on airspeed/altitude, 300 s each) were executed in SIL and HIL for both controllers, from a single script per controller.

1) PID

Table 11 reports the RMS SIL–HIL discrepancy ($t > 12$ s). The residual reaches the quantization floor: pitch attitude within 0.011° in the worst case, altitude within 2.75 m only where the altitude loop is open (integrated drift), and essentially zero elsewhere. Fig. 14 illustrates case 1: the SIL and HIL traces are indistinguishable at plot scale.

2) LQRy

The embedded LQRy is exactly the fixed-gain law of Section III-C compared throughout this article, with the gain set registered in the repository [30] — the bench therefore closes the comparison on the same two builds evaluated in SIL. The LQRy campaign was executed under the same protocol, with per-case firmware builds and scripted programming of the target. Its SIL–HIL residual is larger than the PID's — from 0.23° to 3.57° in pitch depending on the flight case — and is explained by the start-up term quantified in S1:

**FIGURE 14.** PID, flight case 1: SIL vs. HIL on the nonlinear model.**TABLE 12.** Embedded comparison on the bench (HIL, combined mission, identical references, $t > 12$ s).

Channel	PID		LQRy	
	RMS	max	RMS	max
V_T [m/s]	0.07	0.76	0.37	2.11
ψ [°]	1.27	5.79	2.92	10.16
h [m]	1.63	6.82	1.78	10.00

this implementation holds the embedded integrators in reset until the link engages, so the embedded and simulated runs begin from different states. The effect is deterministic and confined to the transient; the doublet responses themselves track closely.

G. CONTROLLER COMPARISON ON THE BENCH

With both controllers on the same bench, the same plant, the same references, and the same metric window, the dominant free variable is the control law — subject to the harness asymmetries disclosed in Section IV-B, which the equalization ablation showed not to drive the conclusions. Table 12 reports the combined mission executed on the hardware: the bench reproduces the SIL verdict of Section IV-E essentially unchanged — the PID tracks airspeed and heading substantially more tightly (RMS 0.07 m/s and 1.27° against 0.37 m/s and 2.92°) with altitude comparable (1.63 versus 1.78 m), and the LQRy's single requirement violation remains the airspeed overshoot (14.6% on the bench against 16.5% in SIL, the small difference owing to the start-up term above). The maxima occur at the reference-step transients. Fig. 15 shows flight case 1.

VII. CONCLUSION

This article covered the full path from controller design to embedded validation for the fixed-wing flight mode of a hybrid VTOL aircraft. A cascade PID and an output-feedback LQR were designed on the linearized model, verified on the nonlinear model, stressed under model mismatch and wind, and deployed to an STM32 microcontroller through automatic code generation, so that the artifact exercised on the bench is the production code itself.

Four results stand out. First, on the nominal plant the

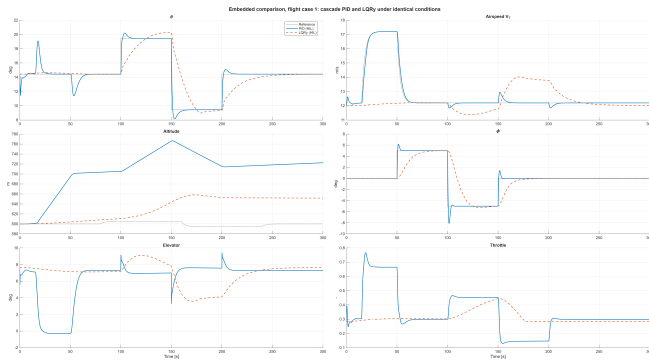


FIGURE 15. Embedded comparison, flight case 1: PID and LQRy under identical conditions on the same bench.

requirements-retuned PID dominates the performance comparison: zero overshoot and the tighter tracking on every guidance channel, meeting every requirement row, while the deliberately soft fixed-gain LQRy concedes a factor of ≈ 5 in airspeed RMS and violates the airspeed-overshoot requirement (16.5%). Second, the robustness study inverts that verdict where it matters most: under a severe, sign-changing lateral-directional model mismatch the PID loses the mission outright, while the fixed-gain LQRy completes it with near-nominal error — a tolerance consistent with the classical robustness heritage of LQ designs, earned here by the deliberately soft tuning; the remaining scenarios add nuance: under the longitudinal-only mismatch the LQRy stays nominal while the PID's clamped altitude loop drifts, under wind and gusts the PID holds every channel more tightly, and both tolerate a plausible +10% identification uncertainty. The comparison thus isolates a genuine performance-robustness trade-off between the two standard laws: which one is preferable depends on whether the dominant threat is model error — where the PID's failure is catastrophic and the LQRy's tolerance is decisive — or atmospheric disturbance, where the PID holds the line more tightly. Third, the lockstep HIL architecture is practically transparent when the campaign is scripted from a single source of references and the controller start-up is synchronized: for the synchronized PID deployment, the SIL-HIL residual reaches the quantization floor of the communication bus (0.011° in pitch attitude in the worst flight case), and the loop is bit-deterministic — the same generated code executed on two different microcontroller models reproduces full 300-second flight cases bit for bit — which makes the bench a measuring instrument and allows a campaign to be reproduced exactly on different hardware. Fourth, exploiting that property, a sensitivity study ranked the implementation choices that actually matter for SIL-HIL agreement: reference consistency between campaigns dominates by two orders of magnitude, start-up desynchronization leaves a persistent residual that link quality cannot remove, low-rate frame loss is benign, and controller discretization — the usual first suspect — is negligible at a 10 ms sample rate.

Future work follows directly from the limitations identified

here: a feedforward decoupling term for the roll-to-pitch coupling quantified in Section III-D, evaluated on both control laws; an extended robustness campaign with stochastic Dryden/von Kármán turbulence [40], [65] in addition to the discrete gusts used here, and with robust syntheses designed for this trade-off [66] as a third contender; equalization of the two design harnesses (actuator models, reference shaping, and protection logic) so that future comparisons are free of these asymmetries; a requirements set rederived specifically for unmanned aircraft in the sense of the scaled flying-qualities literature [46], [59]; extension of the LQRy with gain scheduling over the flight envelope, evaluated under this same SIL-to-HIL protocol; and flight testing of the embedded controllers validated on the bench, closing the simulation-to-flight loop that the HIL stage approximates [60].

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